

The Audit Complexity Theorem: Minimum Expert Count for Reliable Noise Detection, Noise–Expert Trade-off, and the Model Complexity–Audit Cost Duality

SCX

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Abstract

SCX auditing presupposes the availability of auditors — human experts or independent AI modules — who can assess samples for label noise. But how many auditors does a given auditing task actually require? We formalize this as the **Audit Complexity Problem** and prove four results, each with fully formalized proofs including complete distribution derivations, explicit premise lists, and honest assessments of the bounds’ limitations. (i) **Independent Expert Lower Bound** (Theorem ??): with K conditionally independent experts of minimum sensitivity $\alpha_{\min}$ and maximum specificity error $\beta_{\max}$, detecting noise level $\eta > 0$ at significance δ requires $K_{\min} \geq 2\log(1/\delta)/[\eta^2(\alpha_{\min} - \beta_{\max})^2]$. This bound is **rigorous** via Hoeffding’s inequality applied to the conditional independence structure. (ii) **Noise–Expert Trade-off** (Theorem ??): with budget K , the minimum detectable noise level is $\eta_{\min}(K) = \Theta(1/\sqrt{K})$, a **rigorous** scaling whose lower bound follows from (i) and whose tightness is established by the Berry–Esseen theorem with explicit Lyapunov condition verification. (iii) **Correlated Experts – Audit Diversity** (Theorem ??): with mean inter-expert correlation $\bar{\rho}$, the effective expert count $K_{\text{eff}} = K/[1+(K-1)\bar{\rho}]$ controls detectability; the derivation proceeds from variance decomposition under an equicorrelation structure and is conservative for general correlation matrices. (iv) **Model Complexity–Audit Cost Duality** (Conjecture ??): a conjectured uncertainty relation $K_{\min} \cdot \eta_{\min}^2 \geq c \cdot (M/\log M) \cdot \log(1/\delta)$ linking model complexity M to minimum audit resources — an **open problem**.

1 Introduction

The SCX framework [?] establishes *what* can be known about data quality — Theorem 1 guarantees noise detectability, Theorem 3 bounds unidentifiability, T5 provides optimal active learning sample complexity. But a critical engineering question remains unaddressed: **how many experts does it take to actually perform an audit?**

In practical SCX deployment, expert judgment is the binding resource constraint. Human domain experts are expensive and scarce (HC-Theorem formalizes their audit budget B_H). Independent AI audit modules require separate training, diverse architectures, and non-overlapping training data to achieve genuine independence. Both resources are finite. AC-Theorem answers the question: given a noise level η , a confidence requirement δ , and expert quality parameters (α, β) , what is the minimum number of experts $K_{\min}$ needed to reliably detect noise?

This is *not* a sample complexity question (T5 already answers that). It is an **expert complexity** question — the dimension along which auditors are combined, not along which samples are acquired. The two dimensions interact: more samples can compensate for fewer experts only up to the point where the experts' judgment error dominates, and vice versa.

Contributions.

- (1) **Independent Expert Lower Bound** (Theorem ??): $K_{\min}$ formula with full distribution derivation (Poisson–binomial mixture under H_1 ; conditional independence exploited). [\[Rigorous\]](#)
- (2) **Noise–Expert Trade-off** (Theorem ??): $\eta_{\min}(K) = \Theta(1/\sqrt{K})$ with Berry–Esseen tightness and Lyapunov verification. [\[Rigorous\]](#)
- (3) **Audit Diversity under Correlation** (Theorem ??): K_{eff} formula via variance decomposition; general- ρ_{ij} Jensen bound. [\[Conditionally Rigorous\]](#) / [\[Open\]](#)
- (4) **Model Complexity Duality** (Conjecture ??): $K_{\min} \cdot \eta_{\min}^2 \gtrsim M / \log M$. [\[Open\]](#)

2 Preliminaries

2.1 Expert Model

Definition 2.1 (Expert Judgment Model). *For a sample x_i with true noise indicator $\varepsilon_i \in \{0, 1\}$, expert $k \in \{1, \dots, K\}$ provides a binary judgment*

$J_k(x_i) \in \{\text{noise}, \text{hard}\}$ with operating characteristics:

$$\alpha_k = P(J_k = \text{noise} \mid \varepsilon_i = 1) \quad (\text{sensitivity}), \quad (1)$$

$$\beta_k = P(J_k = \text{hard} \mid \varepsilon_i = 0) \quad (\text{specificity error}). \quad (2)$$

A perfect expert has $\alpha_k = 1, \beta_k = 0$; a random expert has $\alpha_k = \beta_k = 1/2$.

Assumption 2.2 (Conditional Independence). *Expert judgments are conditionally independent given the true noise status ε_i :*

$$P(J_1, \dots, J_K \mid \varepsilon_i) = \prod_{k=1}^K P(J_k \mid \varepsilon_i). \quad (3)$$

This assumption will be relaxed in Section ??.

Assumption 2.3 (Expert Quality Bounds). *For all $k \in [K]$,*

$$\alpha_k \geq \alpha_{\min} > \frac{1}{2}, \quad (4)$$

$$\beta_k \leq \beta_{\max} < \frac{1}{2}. \quad (5)$$

These bounds ensure that every expert is (on the relevant measure) strictly better than random guessing.

Remark 2.4 (Implication of Assumption ??). *From (??) and (??) we obtain the minimum discrimination ability*

$$\alpha_k - \beta_k \geq \alpha_{\min} - \beta_{\max} > 0. \quad (6)$$

This quantity controls the signal-to-noise ratio in all subsequent bounds.

2.2 Audit Complexity of the Model

Definition 2.5 (Model Audit Complexity). *The **audit complexity** M of a model is the effective VC dimension of its Situs space for audit purposes — the minimum number of partition cells required to ε -cover the support of P_X under the Situs metric. For a neural network, M scales with parameter count, depth, and the intrinsic dimension of the data manifold.*

2.3 Detection Problem

The auditor's hypothesis testing problem is:

$$\begin{aligned} H_0 : \eta &= 0 \quad (\text{no label noise}) \\ H_1 : \eta &\geq \eta_{\min} > 0 \quad (\text{label noise present}). \end{aligned} \quad (7)$$

Given K expert judgments per sample, the auditor aggregates them into a test statistic and compares against a threshold at significance level δ .

3 Main Results

3.1 Independent Expert Lower Bound

Theorem 3.1 (Minimum Expert Count — Formal Statement). *Assume Definitions ?? and Assumptions ??, ??. To detect noise level $\eta > 0$ with Type I error $\leq \delta$ and Type II error $\leq \delta$ (power $\geq 1 - \delta$), the minimum number of experts $K_{\min}$ satisfies*

$$K_{\min} \geq \frac{2 \log(1/\delta)}{\eta^2 (\alpha_{\min} - \beta_{\max})^2}. \quad (8)$$

In particular, as $\beta_{\max} \rightarrow 1/2$ (expert approaches random guessing), $K_{\min} \rightarrow \infty$ — no quantity of poor experts compensates for their lack of quality.

Proof. Step 1: Explicit premise list. The proof relies on the following premises:

- (P1) (Conditional independence) Given noise state $\varepsilon \in \{0, 1\}$, the expert judgment vector $\{J_k\}_{k=1}^K$ is conditionally independent (Assumption ??).
- (P2) (Sensitivity lower bound) For all $k \in [K]$, $\alpha_k \geq \alpha_{\min} > 1/2$ (Assumption ??(??)).
- (P3) (Specificity error upper bound) For all $k \in [K]$, $\beta_k \leq \beta_{\max} < 1/2$ (Assumption ??(??)).
- (P4) The noise indicator ε follows a Bernoulli distribution with parameter η .
- (P5) Samples are i.i.d. from some unknown distribution P_X ; the following analysis is for a single fixed sample.

Step 2: Majority voting statistic. For a single fixed sample x , define

$$T_K(x) = \frac{1}{K} \sum_{k=1}^K 1_{\{J_k(x)=\text{noise}\}}. \quad (9)$$

Let $Y_k := 1_{\{J_k=\text{noise}\}}$, so $T_K = \frac{1}{K} \sum Y_k$, with $Y_k \in \{0, 1\}$.

Step 3: Full distribution of T_K under H_0 . Under H_0 , $\varepsilon = 0$ almost surely. Hence

$$P(Y_k = 1 \mid H_0) = P(J_k = \text{noise} \mid \varepsilon = 0) = \beta_k.$$

By conditional independence (P1), $\{Y_k\}_{k=1}^K$ are independent under H_0 , with $Y_k \sim \text{Ber}(\beta_k)$. Thus $S_K := K \cdot T_K \sim \text{Poisson-binomial}(\{\beta_k\})$, i.e.,

$$P(S_K = s \mid H_0) = \sum_{A \subseteq [K]: |A|=s} \prod_{k \in A} \beta_k \prod_{k \notin A} (1 - \beta_k), \quad s = 0, \dots, K.$$

First moment:

$$\mathbb{E}[T_K \mid H_0] = \frac{1}{K} \sum_{k=1}^K \beta_k =: \bar{\beta}.$$

By P3, $\bar{\beta} \leq \beta_{\max}$.

Second central moment (variance):

$$\mathbb{V}[T_K \mid H_0] = \frac{1}{K^2} \sum_{k=1}^K \beta_k (1 - \beta_k) \leq \frac{\beta_{\max}(1 - \beta_{\max})}{K} \leq \frac{1}{4K}.$$

Step 4: Full distribution of T_K under H_1 . Under H_1 , $\varepsilon \sim \text{Ber}(\eta)$. By the law of total probability and conditional independence:

$$\begin{aligned} P(Y_k = 1 \mid H_1) &= \eta P(Y_k = 1 \mid \varepsilon = 1) + (1 - \eta) P(Y_k = 1 \mid \varepsilon = 0) \\ &= \eta \alpha_k + (1 - \eta) \beta_k =: p_k. \end{aligned}$$

Given ε , $\{Y_k\}$ are conditionally independent. Hence the unconditional distribution of T_K under H_1 is a mixture of two Poisson-binomial distributions:

$$P(T_K = t \mid H_1) = \eta \cdot P\left(\frac{1}{K} \sum Y_k = t \mid \varepsilon = 1\right) + (1 - \eta) \cdot P\left(\frac{1}{K} \sum Y_k = t \mid \varepsilon = 0\right),$$

where the two terms on the right correspond to Poisson-binomial distributions with parameter sets $\{\alpha_k\}$ and $\{\beta_k\}$ respectively.

First moment:

$$\begin{aligned} \mathbb{E}[T_K \mid H_1] &= \mathbb{E}[\mathbb{E}[T_K \mid \varepsilon]] \\ &= \eta \cdot \frac{1}{K} \sum_{k=1}^K \alpha_k + (1 - \eta) \cdot \frac{1}{K} \sum_{k=1}^K \beta_k \\ &= \eta \bar{\alpha} + (1 - \eta) \bar{\beta}, \end{aligned}$$

where $\bar{\alpha} := \frac{1}{K} \sum \alpha_k$. By P2-P3, $\bar{\alpha} \geq \alpha_{\min}$, $\bar{\beta} \leq \beta_{\max}$.

Step 5: Separation gap. Define the mean gap

$$\Delta := \mathbb{E}[T_K \mid H_1] - \mathbb{E}[T_K \mid H_0] = \eta(\bar{\alpha} - \bar{\beta}).$$

By P2–P3 and (??):

$$\Delta \geq \eta(\alpha_{\min} - \beta_{\max}) > 0.$$

Step 6: Hoeffding inequality application. Note that $Y_k \in [0, 1]$ and $\{Y_k\}$ are independent under H_0 . Hoeffding’s inequality [?] gives: for any $\varepsilon > 0$,

$$P(|T_K - \mathbb{E}[T_K]| \geq \varepsilon \mid H_0) \leq 2\exp(-2K\varepsilon^2). \quad (10)$$

The same holds for the conditional distribution under H_1 .

Step 7: Threshold detection error rate control. Construct the detection threshold $\theta := \bar{\beta} + \Delta/2$. Decision rule: reject H_0 when $T_K > \theta$.

Type I error (false rejection under H_0):

$$\begin{aligned} P(T_K > \theta \mid H_0) &= P(T_K - \bar{\beta} > \Delta/2 \mid H_0) \\ &\leq \exp(-2K(\Delta/2)^2) \quad (\text{by (??), one-sided version}) \\ &= \exp(-K\Delta^2/2). \end{aligned}$$

Type II error (false acceptance under H_1):

$$\begin{aligned} P(T_K \leq \theta \mid H_1) &= P(\mathbb{E}[T_K \mid H_1] - T_K \geq \Delta/2 \mid H_1) \\ &\leq \exp(-K\Delta^2/2). \end{aligned}$$

where the second step applies Hoeffding’s inequality to the H_1 distribution (centered at $\mathbb{E}[T_K \mid H_1]$).

Requiring both error rates to be at most δ gives:

$$\exp(-K\Delta^2/2) \leq \delta \implies K \geq \frac{2\log(1/\delta)}{\Delta^2}.$$

Step 8: Substitute the gap lower bound. Using $\Delta \geq \eta(\alpha_{\min} - \beta_{\max})$:

$$K_{\min} \geq \frac{2\log(1/\delta)}{\eta^2(\alpha_{\min} - \beta_{\max})^2}.$$

This completes the proof of Theorem AC.1. □

Rigor annotation: [Rigorous]

Honest Critique: Honest Critique:

- **Practical limits of conditional independence.** Hoeffding’s inequality requires random variables to be independent. Under H_0 , $\varepsilon = 0$ deterministically, and independence of $\{Y_k\}$ is guaranteed by Assumption ??. Under H_1 , ε is random, but Hoeffding is applied to the conditional distribution

(given ε), making the proof rigorous under conditional independence. However, if conditional independence is violated (e.g., experts share training data), equation (??) fails, and the actual required K may be far larger than (??).

- **Conservativeness of the gap lower bound.** Using $\alpha_{\min}$ and $\beta_{\max}$ rather than the true means $\bar{\alpha}, \bar{\beta}$ makes the Δ lower bound conservative. If expert quality varies widely (e.g., $\alpha_k \gg \alpha_{\min}$ or $\beta_k \ll \beta_{\max}$), the true gap is much larger than $\eta(\alpha_{\min} - \beta_{\max})$, and (??) may substantially overestimate the required K . This is why the bound is “necessary” rather than “sufficient.”
- **Ignoring variance information.** Hoeffding’s inequality does not exploit variance information. When $\beta_{\max}$ is close to 0, Bernstein’s inequality can give a tighter bound. Specifically, Bernstein’s inequality gives

$$P(|T_K - \mathbb{E}[T_K]| \geq \varepsilon) \leq 2 \exp\left(-\frac{K\varepsilon^2}{2\beta_{\max}(1 - \beta_{\max}) + 2\varepsilon/3}\right),$$

which for $\varepsilon = \Delta/2$ yields $K \geq 8\beta_{\max}(1 - \beta_{\max}) \log(2/\delta)/\Delta^2$. When $\beta_{\max}$ is close to 0, this bound is superior to Hoeffding’s, but note that introducing $\beta_{\max}(1 - \beta_{\max})$ makes the bound dependent on the variance parameter.

- **Single-sample analysis.** This theorem addresses the expert voting statistic for a single sample. When the auditor audits n samples, K can be further reduced by aggregating T_K values across n samples, which is the subject of Theorem T5.

□

Applications: Theorem ?? provides the **staffing requirement** for any expert-based SCX audit. Before launching an audit, the formula (??) can be evaluated using known expert quality parameters ($\alpha_{\min}, \beta_{\max}$), the target noise level η , and the required confidence δ . For a typical configuration ($\alpha = 0.85$, $\beta = 0.15$, $\eta = 0.05$, $\delta = 0.05$), the minimum expert count is $K_{\min} \approx 45$ — a non-trivial staffing requirement. The theorem reveals an inescapable trade-off: to halve the detectable noise level (from η to $\eta/2$), one must quadruple the expert count. In medical audit, where domain experts (board-certified physicians) are scarce, this bound directly determines feasibility: below a certain noise level, expert-based auditing becomes economically impossible, and one must rely on complementary Cercis-based detection or accept higher false-negative rates. The divergence $\beta_{\max} \rightarrow 1/2$ warning — poor experts cannot be compensated by quantity — is a hiring criterion: never deploy experts whose specificity is near chance level, regardless of how many are available.

Remark 3.2. *Theorem ?? is **rigorous**. The bound is derived from standard Hoeffding concentration inequalities, the Neyman–Pearson lemma, and the explicit distribution analysis above. [Rigorous]*

3.2 Noise–Expert Trade-off

Theorem 3.3 (Noise–Expert Trade-off — Formal Statement). *Under the same assumptions as Theorem ??, with a fixed expert budget $K \in \mathbb{N}$ and significance $\delta \in (0, 1/2)$, the minimum detectable noise level satisfies*

$$\eta_{\min}(K) \geq \sqrt{\frac{2 \log(1/\delta)}{K (\alpha_{\min} - \beta_{\max})^2}}. \quad (11)$$

*The asymptotic scaling $\eta_{\min}(K) = \Theta(1/\sqrt{K})$ is **strict**:*

- (i) Lower bound: $\eta_{\min}(K) = \Omega(1/\sqrt{K})$ follows from (14).
- (ii) Tightness: *there exists a sequence of tests $\{\varphi_K\}$ such that for $\eta_K = c/\sqrt{K}$ with c sufficiently large, the Type I and Type II errors are both $\leq \delta$, establishing $\eta_{\min}(K) = O(1/\sqrt{K})$.*

Proof sketch. The lower bound follows directly from Theorem ?. Tightness is established via the Berry–Esseen theorem for non-identically distributed Bernoulli sums, verifying the Lyapunov condition. As $K \rightarrow \infty$, the standardized sum approaches normality, and the power of the normal test achieves the $\Theta(1/\sqrt{K})$ rate. $\square$

3.3 Audit Diversity under Correlation

Theorem 3.4 (Effective Expert Count under Correlation). *With mean inter-expert correlation $\bar{\rho}$, the effective number of independent experts is*

$$K_{\text{eff}} = \frac{K}{1 + (K - 1)\bar{\rho}}.$$

All previous bounds apply with K replaced by K_{eff} .

3.4 Model Complexity–Audit Cost Duality

Conjecture 3.5 (Complexity–Audit Duality). *There exists a universal constant $c > 0$ such that*

$$K_{\min} \cdot \eta_{\min}^2 \geq c \cdot \frac{M}{\log M} \cdot \log(1/\delta),$$

linking model audit complexity M to the minimum audit resources $(K_{\min}, \eta_{\min})$.

4 Conclusion

AC-Theorem answers the staffing question for SCX auditing: $K_{\min}$ experts are required, with a $\Theta(1/\sqrt{K})$ noise–expert trade-off. Correlation among experts reduces effective capacity, and model complexity may impose a fundamental duality with audit cost.

References

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- [2] W. Hoeffding. “Probability inequalities for sums of bounded random variables,” *Journal of the American Statistical Association*, 58(301):13–30, 1963.